

GRADE 420F - 420 STAINLESS STEEL

420F stainless steel is a martensitic grade of stainless. It is magnetic and hardenable with rust resistant properties. 420F stainless is characterised by having a 0.28-0.38 carbon, and 15-17% chromium content. This mix of analysis gives 420F stainless steel an outstanding corrosion resistance and exceptional wear properties after heat treatment. Superior to 410 and 416 type stainlesses for most applications. 420F stainless steel has good ductility in the annealed condition, but is capable of achieving 55HRC after treatment. Optimal corrosion resistance is realised when 420F is surface ground and polished after hardening.

420F - 420 Stainless Steel Chemical Analysis wt %						
C	Si	Mn	P	S	Cr	Ni
0.28-0.38	≤ 1.00	≤ 1.40	≤ 0.03	0.05-0.10	15.0-17.0	≤ 1.00

APPLICATIONS FOR 420F

Dental and surgical instruments cutlery, pump components, shafts, plastic moulds, gear pinions, cams and various hand tools

SOFT ANNEALING 420F

Heat to 850-880°C. Cool slowly in air.

MACHINEABILITY OF 420F

Good machining characteristics due to the addition of Sulphur.

420F - 420 Stainless Mechanical Properties at ambient temperature

Condition annealed. Quenched and tempered 850°C

0.2% Proof Stress: 905 Nmm²

Tensile Strength: 950 - 1100 Nmm²

Elongation: 10%

Notch Impact Energy: (ISO-V) (J) 20

Hardness HB Max 230

FORGING 420F

Hot forming temperature 800-1100°C. Slow cool

HARDENING 420F

Harden from a temperature 1000-1050°C

Follow by oil or air quenching.

TEMPERING 420F

Tempering temperature 150-200°C

HEAT TREATMENT ADVISE OF 420F

Treatment temperatures, including the rate of heating, cooling and soaking times will vary due to factors such as the shape and size of each 420 stainless steel component. Other considerations during the heat treatment process include the type of furnace, quenching medium, and workpiece transfer facilities. Please consult your heat treatment provider for full guidance for heat treatment of 420F stainless steel.

CORROSION RESISTANCE 420F

Hardened condition realises resistance to atmosphere, fresh, dilute acids and alkalise, fresh water, fruit juices, vegetable juices and plastics

420F - 420 Stainless Steel Physical properties

Average values at ambient temperature

Modulus of elasticity. ($10^3 \times \text{N/mm}^2$): 216

Density. (g/cm^3): 7.73

Thermal conductivity. (W/m.K): 31.0

Electric resistivity. ($\text{Ohm mm}^2/\text{m}$): 0.60

Specific heat capacity. (J/g.K): 0.47

Magnetisable. Yes

WELDING OF 420F

Not commonly welded due to its air-hardening characteristics. Welding may possibly be performed after preheating to 149-204°C with the post-welding tempering at temperature for 2 hours. Recommendation of specialist advice to be sought

COLD WORKING OF 420F

The material will withstand only minor cold work. Radical forming operations will result in cracking

Disclaimer

The information and data presented herein are typical or average values and are not a guarantee of maximum or minimum values. Applications specifically suggested for material described herein are made solely for the purpose of illustrations to enable the reader to make their own evaluations. They are not intended to either express or imply the suitability for these or any other purposes

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